Introduction to Biophysics II: Quantum Biology

Lecture 04
Absorption and emission of light

- Absorption and stimulated emission
- Einstein coefficients and spontaneous emission
- Emission line width

1

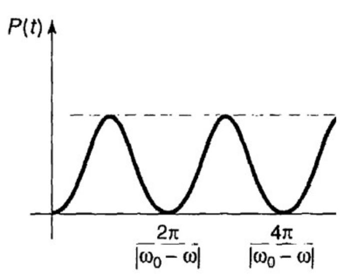
Transition between states

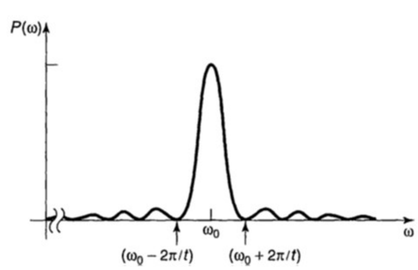
Harmonic interaction: $H'_{ab} = V_{ab}\cos(\omega t)$ $\quad V_{ab} \equiv \langle \psi_a | V | \psi_b \rangle$

Transition probability:

$$\boxed{P_{a\to b}(t) = |c_b(t)|^2 \cong \frac{|V_{ab}|^2}{\hbar^2}\frac{\sin^2[(\omega_0-\omega)t/2]}{(\omega_0-\omega)^2}}.$$

The probability that particle that was initially in state ψ_a ends up in state ψ_b at time t.





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Transition between states

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Transition probability:

$$\boxed{P_{a\to b}(t) = |c_b(t)|^2 \cong \frac{|V_{ab}|^2}{\hbar^2} \frac{\sin^2[(\omega_0 - \omega)t/2]}{(\omega_0 - \omega)^2}}$$

A few notes about V_{ab}:

Complete orthogonal basis wavefunctions are ψ_a and ψ_b

For $V_{ab} \equiv \langle \psi_a | V | \psi_b \rangle$ to be nonzero the $\hat{V}\psi_b$ must modify the wavefunction so that it will have some of the ψ_a character in it, i.e.:

$$\hat{V}\psi_b = d_1\psi_b + d_2\psi_a$$

$$\left\langle \psi_a | \hat{V} | \psi_b \right\rangle = \left\langle \psi_a | d_1\psi_b + d_2\psi_a \right\rangle = d_2$$

→ d_2 should not be zero!

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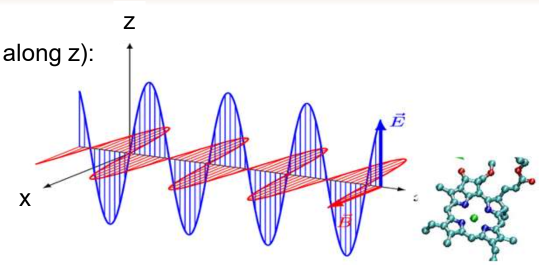
Absorption/emission of light

Electric field
(assume monochromatic, polarized along z):

$$\mathbf{E} = E_0 \cos(\omega t)\,\hat{k}$$



Energy (work) for moving charge q in electric field along z:

$$\int_{r\_from}^{r\_to} \vec{F}_e \cdot d\vec{r} = \int_{r\_from}^{r\_to} q\vec{E} \cdot d\vec{r} = qE\left(z_{to} - z_{from}\right) = qEz$$

Potential energy is '-' work done: $H' = -qE_0 z \cos(\omega t)$

Where q is charge of an electron

Notes:
- nucleus is heavy (10^4 heavier than electron), ignore its motion
- ψ is even or odd function of z, diagonal elements of H' vanish: $z|\psi|^2$ is odd:

$$H'_{aa} \equiv \left\langle \psi_a | -qEz | \psi_a \right\rangle = -qE\int \left|\psi_a\right|^2 z dz = 0$$

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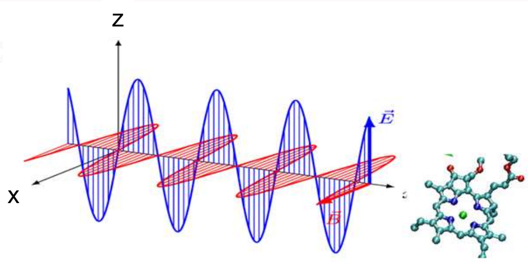
Absorption/emission of light

Electric field
(assume monochromatic, polarized):

$$\mathbf{E} = E_0 \cos(\omega t)\,\hat{k}$$

$$H' = -qE_0 z \cos(\omega t)$$



$$H'_{ab} = V_{ab}\cos(\omega t)$$

$$V_{ab} \equiv \langle \psi_a | V | \psi_b \rangle$$

Polarized along *z*

$$H'_{ba} = -\wp E_0 \cos(\omega t), \quad \text{where } \wp \equiv q\langle \psi_b | z | \psi_a \rangle$$

p = off-diagonal element of dipole moment operator, Similar to *electric dipole* in electrostatics
But: t is associated with *transition dipole. This is z-component of dipole moment operator.*
z is the distance between coordinates of ψ_a and ψ_b, in z-direction $z=(z_b-z_a)$, we move electron density from one distribution to another.

And the equation we derived earlier works with $V_{ba} = -\wp E_0$

$$\boxed{P_{a\to b}(t) = \left(\frac{|\wp|E_0}{\hbar}\right)^2 \frac{\sin^2[(\omega_0-\omega)t/2]}{(\omega_0-\omega)^2}}$$

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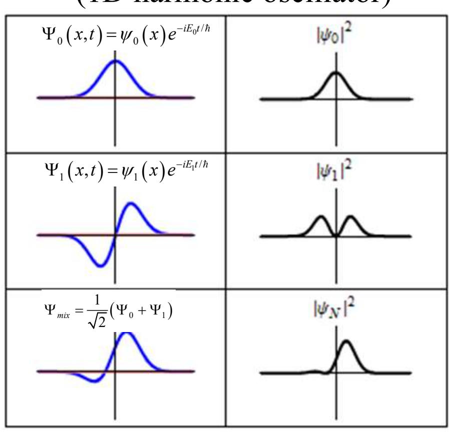
Notes on $\rho = q\left\langle \psi_b \middle| z \middle| \psi_a \right\rangle$

(1D harmonic oscillator)

$$\psi_0 = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar}x^2}$$

$$\psi_1 = \sqrt{\frac{2m\omega}{\hbar}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4} x e^{-\frac{m\omega}{2\hbar}x^2}$$

Mixed state:



$$|\Psi_0|^2 \sim e^{-\frac{m\omega}{\hbar}x^2}$$

$$|\Psi_1|^2 \sim x^2 e^{-\frac{m\omega}{\hbar}x^2}$$

Oscillating electric field perturbs the ground state ‘stationary’ electron density distribution adding an oscillatory component at electric field frequency. The new state will thus have a character of a mixed ground+excited state. As the result it will be a superposition of both states and there will be probability to find the molecule in excited state.

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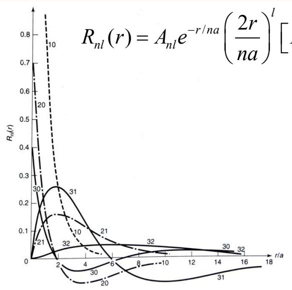
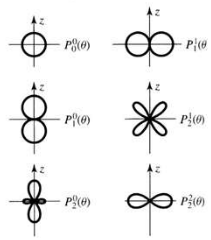
Example: s,p,d orbitals

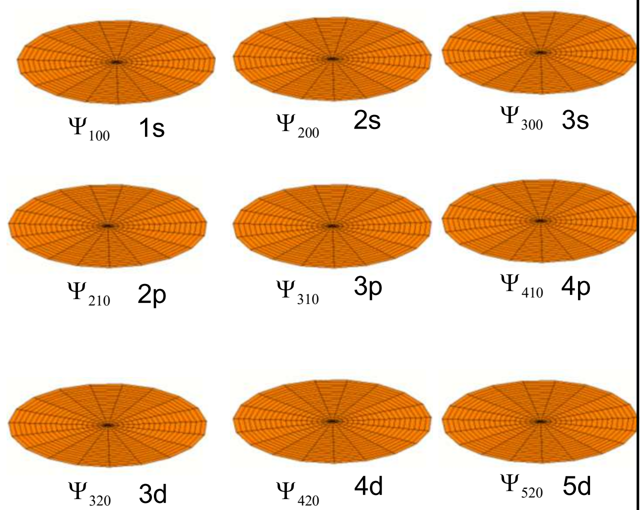
$n = 1, 2, 3, \ldots$
$l = 0, 1, 2, \ldots, n-1$
$m = -l, \ldots, 0, \ldots, +l$

$$R_{nl}(r) = A_{nl} e^{-r/na} \left(\frac{2r}{na}\right)^l \left[L_{n-l-1}^{2l+1}(2r/na)\right]$$



$$\Theta(\theta) = A P_l^m(\cos\theta)$$





http://en.wikipedia.org/wiki/Atomic_orbital

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Absorption/emission of light

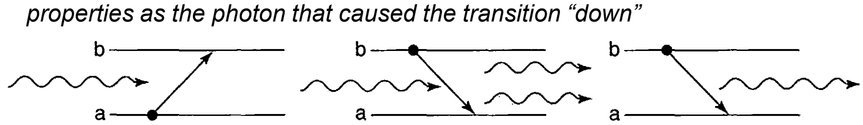
$$\boxed{P_{a\to b}(t) = \left(\frac{|\wp|E_0}{\hbar}\right)^2 \frac{\sin^2[(\omega_0-\omega)t/2]}{(\omega_0-\omega)^2}} \qquad \omega_0 \equiv \frac{E_b - E_a}{\hbar}$$

Note:
- electromagnetic field of frequency ω consists of *photons* with energy $E_{ph}=\hbar\omega$
- Energy difference between states *a* and *b* is also E_b-E_a=$\hbar\omega_0$

  → energy is *conserved – one photon is absorbed per one transition*

Note:
- The derivation was done in assumption that $E_b > E_a$ for transition *a→b*
- Can *swap indices* and repeat derivation for transition from *b→a*, the probability will be exactly the same (for 2-level system)

This was first described by Einstein: stimulated emission. Instead of absorbing a photon, one additional photon will be emitted with exact same properties as the photon that caused the transition "down"



(a) Absorption (b) Stimulated emission (c) Spontaneous emission

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Absorption/emission of light

Incoherent perturbation

Start from coherent: $$P_{a\to b}(t) = \left(\frac{|\wp|E_0}{\hbar}\right)^2 \frac{\sin^2[(\omega_0-\omega)t/2]}{(\omega_0-\omega)^2}$$

Energy density of electric field: $u = \frac{\epsilon_0}{2}E_0^2.$ *Permittivity of free space*

$$P_{b\to a}(t) = \frac{2u}{\epsilon_0\hbar^2}|\wp|^2 \frac{\sin^2[(\omega_0-\omega)t/2]}{(\omega_0-\omega)^2}$$

The transition probability is proportional to the electric field energy density

Assume light is a mixture of different frequencies with a spectrum $\rho(\omega)$
i.e. $\rho(\omega)d\omega$ is the energy density at frequency range $d\omega$ around ω

$$P_{b\to a}(t) = \frac{2}{\epsilon_0\hbar^2}|\wp|^2 \int_0^\infty \rho(\omega)\left\{\frac{\sin^2[(\omega_0-\omega)t/2]}{(\omega_0-\omega)^2}\right\} d\omega.$$

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Absorption/emission of light

Incoherent perturbation

$$P_{b\to a}(t) = \frac{2}{\epsilon_0\hbar^2}|\wp|^2 \int_0^\infty \rho(\omega)\left\{\frac{\sin^2[(\omega_0-\omega)t/2]}{(\omega_0-\omega)^2}\right\} d\omega.$$

Term in {…} sharply peaks at ω_0, replace $\rho(\omega)=\rho(\omega_0)$:

$$P_{b\to a}(t) \cong \frac{2|\wp|^2}{\epsilon_0\hbar^2}\rho(\omega_0) \int_0^\infty \frac{\sin^2[(\omega_0-\omega)t/2]}{(\omega_0-\omega)^2} d\omega$$

This is table integral:

$$P_{b\to a}(t) \cong \frac{\pi|\wp|^2}{\epsilon_0\hbar^2}\rho(\omega_0)t$$

Transition is *proportional* to *t*

Transition rate: $$R_{b\to a} = \frac{\pi}{\epsilon_0\hbar^2}|\wp|^2\rho(\omega_0)$$

$$R \equiv dP/dt.$$

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Absorption/emission of light

Incoherent perturbation

$$R_{b\to a} = \frac{\pi}{\epsilon_0 \hbar^2} |\wp|^2 \rho(\omega_0)$$

Assumed that: light comes along x, polarized in *z* direction.
→ Need to average over all directions.

replace $|\wp|^2$ with the average of $|\boldsymbol{\wp} \cdot \hat{n}|^2$, where $\boldsymbol{\wp} \equiv q\langle\psi_b|\mathbf{r}|\psi_a\rangle$

Transition dipole along *z*

Polarization of light

Transition dipole vector

Transition rate: $\boxed{R_{b\to a} = \frac{\pi}{3\epsilon_0 \hbar^2} |\boldsymbol{\wp}|^2 \rho(\omega_0)}$ $\quad R \equiv dP/dt$

3: averaged over all possible directions

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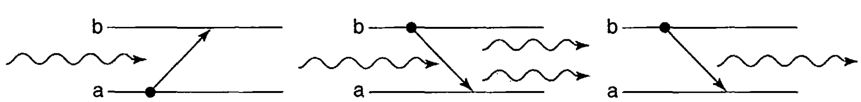
Absorption/emission of light



(a) Absorption (b) Stimulated emission (c) Spontaneous emission

Spontaneous emission: How is it possible????

Quantum electrodynamics: fields are not zero even in ground state, there is always some perturbation by the ground state electric field.

We will not discuss quantum electrodynamics…

But Einstein gave insight to lifetimes of spontaneous emission without quantum electrodynamics!

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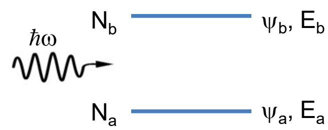
Absorption/emission of light

Einstein A and B coefficients



Given: ensemble of two-state molecules
Let *A* be a spontaneous emission rate

$$\frac{dN_b}{dt} = -N_b A - N_b B_{ba}\rho(\omega_0) + N_a B_{ab}\rho(\omega_0)$$

Recall: $R_{b\to a} = \boxed{\frac{\pi}{\epsilon_0 \hbar^2}|\wp|^2}\rho(\omega_0)$

Assume: dynamic equilibrium with the ambient EM field, $dN_b/dt=0$ →

$$\rho(\omega_0) = \frac{A}{(N_a/N_b)B_{ab} - B_{ba}}$$

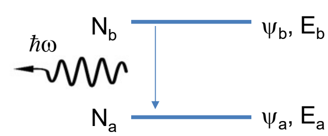
$$\frac{N_a}{N_b} = \frac{e^{-E_a/k_B T}}{e^{-E_b/k_B T}} = e^{\hbar\omega_0/k_B T}$$

It must be in Boltzmann equilibrium!

$$\rho(\omega_0) = \frac{A}{e^{\hbar\omega_0/k_B T} B_{ab} - B_{ba}}$$

Absorption/emission of light

Einstein A and B coefficients



$$\rho(\omega_0) = \frac{A}{e^{\hbar\omega_0/k_B T} B_{ab} - B_{ba}}$$

Recall Planck's blackbody formula:

$$\rho(\omega) = \frac{\hbar}{\pi^2 c^3}\frac{\omega^3}{e^{\hbar\omega/k_B T} - 1}.$$

What does that mean?

$$B_{ab} = B_{ba}$$

Einstein was *forced* to invent stimulated emission to derive Planck's formula!

$$\rho(\omega_0) = \frac{1}{B_{ba}}\frac{A}{e^{\hbar\omega_0/k_B T} - 1}$$

Must match!

$$A = \frac{\omega_0^3 \hbar}{\pi^2 c^3} B_{ba}$$

$$B_{ba} = \frac{\pi}{3\epsilon_0 \hbar^2}|\wp|^2.$$

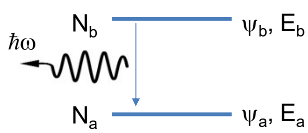
Spontaneous emission rate

$$\boxed{A = \frac{\omega_0^3 |\wp|^2}{3\pi\epsilon_0 \hbar c^3}}$$

Lifetime of Excited State

$$A = \frac{\omega_0^3 |\boldsymbol{\wp}|^2}{3\pi\epsilon_0 \hbar c^3}.$$

$$\boxed{\frac{dN_b}{dt}} = -N_b A - N_b B_{ba}\rho(\omega_0) + N_a B_{ab}\rho(\omega_0)$$



$$\Downarrow$$

$$N_b(t) = N_b(0)e^{-At}$$

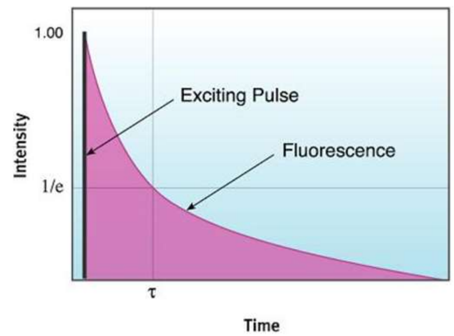
Decay time: $\tau = \frac{1}{A}$

Usually decay rate is denoted as *k:*
$k = 1/\tau$

Note: if there are more than one decay channel with rates k_i, rates add up:

Net rate: $k = k_1 + k_2 + \dots$

Net decay time: $1/\tau = 1/\tau_1 + 1/\tau_2 + \dots$



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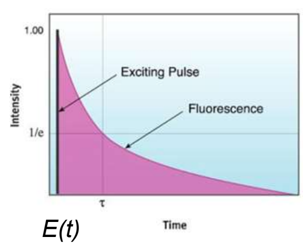
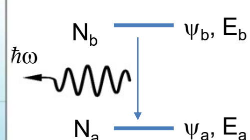
Fluorescence Linewidth

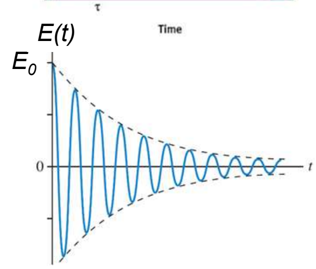
$$N_b(t) = N_b(0)e^{-kt} = N_b(0)e^{-\frac{t}{\tau}}$$





$$\Downarrow$$

$$E(t) = E_0 \exp\left(-\frac{t}{\tau}\right)\cos(\omega t)$$



What will be the actual spectrum of that EM wave?

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Recall: Fourier theorem

Any periodic function $f(t)$ with period T can be expanded as:

$$f(t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\left[a_n \cos(n\omega t) + b_n \sin(n\omega t)\right] \qquad \omega \equiv \frac{2\pi}{T}$$

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\cos(n\omega t)\,dt \qquad n = 0, 1, 2, \ldots$$

$$b_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\sin(n\omega t)\,dt \qquad n = 1, 2, \ldots$$

Alternatively:

$$f(t) = \sum_{n=1}^{\infty} c_n e^{in\omega t} \qquad c_n = \frac{1}{T}\int_{-T/2}^{T/2} f(t)e^{-in\omega t}\,dt$$

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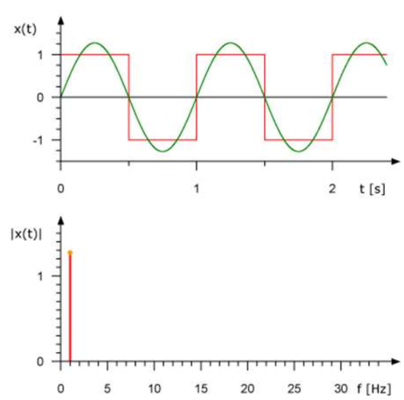
Fourie composition: example



https://youtu.be/LznjC4Lo7lE

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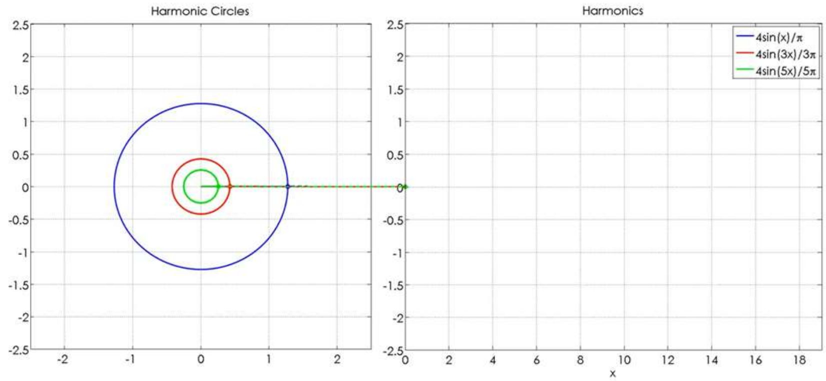
Recall: Fourier theorem



Create your own: https://www.geogebra.org/m/aenhzjpz

Cool visualization: https://www.youtube.com/watch?v=r6sGWTCMz2k

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Fluorescence Linewidth

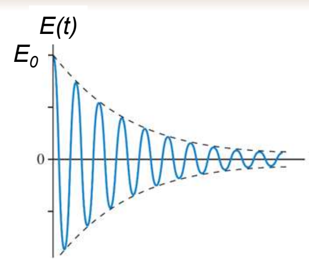
$$E(t) = E_0 \exp\left(-\frac{t}{\tau}\right)\cos(\omega t)$$

Or: $E(t) = E_0 \cdot e^{-\gamma t} \cdot e^{i\omega_0 t} +$ C.C.

→ Fourier transform: $\gamma \equiv 1/\tau$

$$E(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} a(\omega) e^{i\omega t} d\omega$$

$$a(\omega) = \int_{-\infty}^{+\infty} E(t) e^{-i\omega t} dt$$



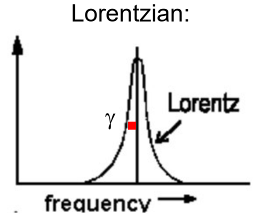
Animated: https://youtu.be/xBxJEc1yTUc

Solution:

$$a(\omega) = -E\left\{\frac{1}{i(\omega_0-\omega)-\gamma} - \frac{1}{i(\omega_0+\omega)+\gamma}\right\}$$

$$I(\omega) = -E_0^2 \cdot \frac{1}{(\omega-\omega_0)^2+\gamma^2}$$

Lorentzian:



This is “natural linewidth”: $\Delta\omega = 2\gamma \equiv 2/\tau$

full width at half maximum, **fwhm**

http://www.pci.tu-bs.de/aggericke/PC4e/Kap_III/Linienbreite.htm

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Lifetime & Uncertainty Principle

Heisenberg energy-time uncertainty principle:

$$\Delta E \Delta t \geq \hbar / 2$$

Meaning: A state that only exists for a short time cannot have a definite energy.

$$\Delta E \geq \frac{\hbar}{2\tau}$$

Also: $\Delta E = \hbar \Delta \omega$

$$\omega_0 \equiv \frac{E_b - E_a}{\hbar}$$

$$\Delta \omega \geq \frac{1}{2\tau}$$

Compare with $\Delta \omega = 2\gamma \equiv 2 / \tau$

rms value of frequency

Fwhm of lorentzian

Same order of magnitude, exact number depends on definition of width $\Delta\omega$

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Lifetime & Uncertainty Principle

$$\Delta \omega = 2\gamma \equiv 2 / \tau$$

Example:

Lifetime of chlorophyll *a* (Chl *a*) is ~5 ns.
Its absorption maximum is at 670 nm

What is its natural linewidth?

$$\Delta \nu = \frac{\Delta \omega}{2\pi} = \frac{1}{\pi \tau} = \frac{1}{\pi 5 \cdot 10^{-9} s} = 6.4 \cdot 10^{7}\, Hz = 0.0021 cm^{-1}$$

$$\Delta \lambda \approx \frac{\lambda^2}{c} \Delta \nu = 1 \cdot 10^{-4}\, nm$$

Convert to nm:

$$\nu = \frac{c}{\lambda}$$

$$d\nu = \frac{c}{\lambda^2} d\lambda$$

$$d\lambda = \frac{\lambda^2}{c} d\nu$$

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Lifetime & Absorption bandwidth

Steady state spectroscopy *can* be used to measure lifetime of excited state

$$\Delta\nu = \frac{1}{\pi\tau} \quad \Rightarrow \quad \tau = \frac{1}{\pi\Delta\nu}$$

Can rewrite for ps (1 picosecond = 10^{-12}s) and cm^{-1} (wavenumbers)

$$\tau(ps)\cdot 10^{-12}\frac{s}{ps} = \frac{1}{\pi\Delta\nu\left(cm^{-1}\right)3\cdot 10^{10}\left(\frac{cm}{s}\right)}$$

$$\tau(ps) = \frac{10.6\left(ps\cdot cm^{-1}\right)}{\Delta\nu\left(cm^{-1}\right)}$$

Or for product (uncertainty):
$\tau(ps)\Delta\nu\left(cm^{-1}\right) = 10.6(ps\cdot cm)$

lifetime	fwhm
10 fs	1060 cm^{-1}
100 fs	106 cm^{-1}
1 ps	10.6 cm^{-1}
10 ps	1.06 cm^{-1}
100 ps	0.106 cm^{-1}
1ns	0.011 cm^{-1}

Note: this is *natural* linewidth defined by lifetime of state only

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Linewidth broadening

Example:

Lifetime of chlorophyll *a* (Chl *a*) is ~5 ns.
Its absorption maximum is at 670 nm

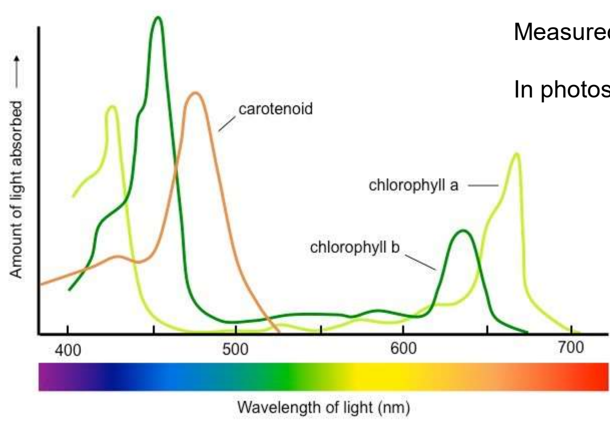
natural linewidth:

$$\Delta\nu = 6.4\cdot 10^{7}\,Hz = 0.0021 cm^{-1}$$

$$\Delta\lambda \approx 1\cdot 10^{-4}\,nm$$

Measured in solution: ~7 nm

In photosystem I : ~30 nm



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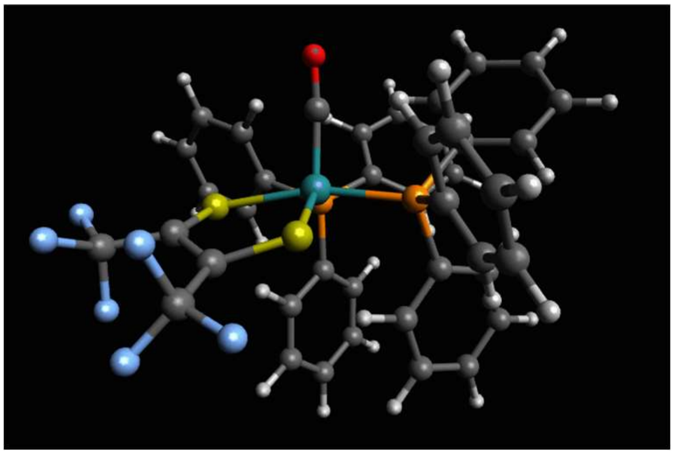
Linewidth broadening

Molecular vibrations:



T M Porter et al, Chem. Sci., 2018, DOI: 10.1039/c8sc03258k

https://www.chemistryworld.com/news/molecular-shapeshifting-observed-at-ultrafast-speeds/3009778.article

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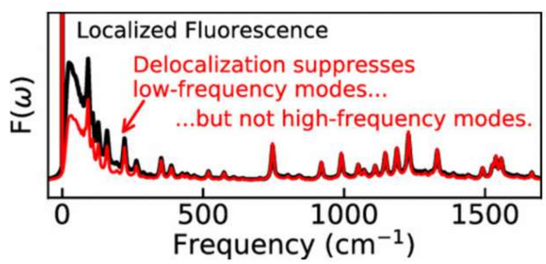
Linewidth broadening

Molecular vibrations:

Example: chlorophyll



Mike Reppert: . Phys. Chem. B 2020, 124, 45, 10024–10033

Molecular vibrations have discrete energies
Some vibrational modes are coupled to electronic transitions
Along with a photon, a phonon could be emitted or absorbed resulting in additional line in spectrum
Can measure using spectroscopy (Raman spectroscopy) and model using known structure

https://pubs.acs.org/doi/abs/10.1021/acs.jpcb.0c05789

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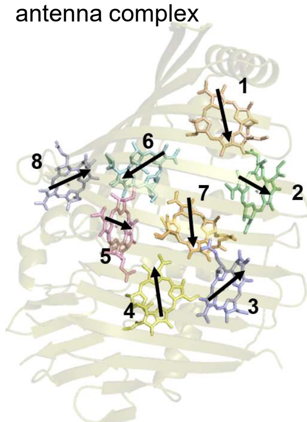
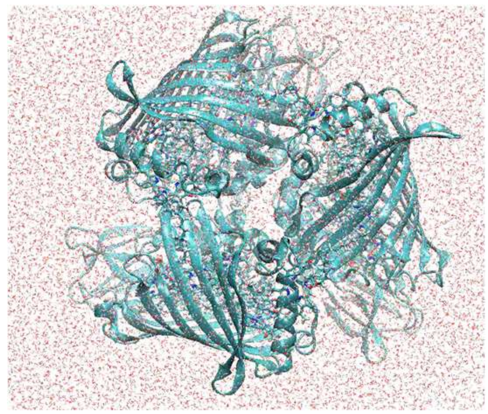
Linewidth broadening

Coupling with "bath": molecule is embedded into protein which has its own vibrational modes, acoustic modes are low-frequency almost continuous in spectrum

Example: FMO
Fenna Mathews Olson
antenna complex



FMO trimer molecular dynamics at room T
(Yongbin Kim, Purdue)



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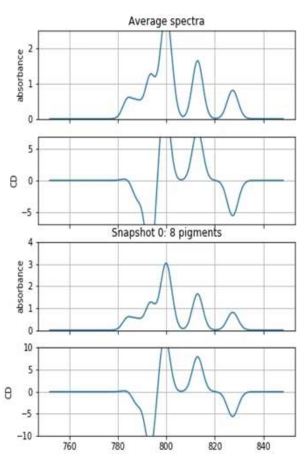
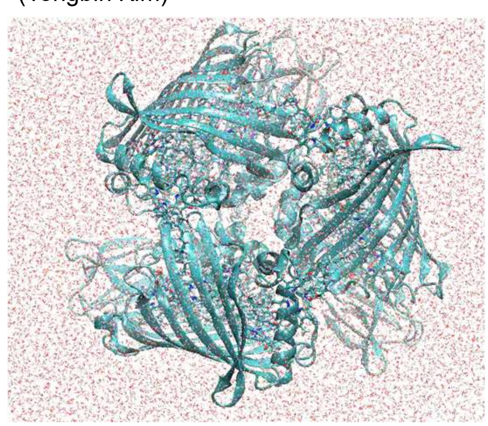
Linewidth broadening

Coupling with "bath": molecule is embedded into protein which has its own vibrational modes, acoustic modes are low-frequency almost continuous in spectrum



FMO trimer molecular dynamics at room T
(Yongbin Kim)



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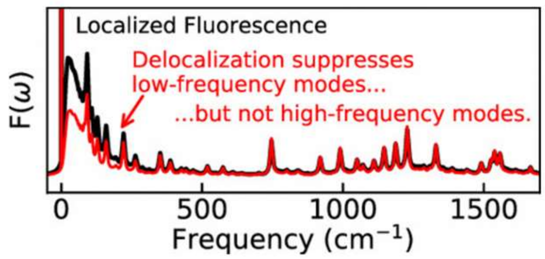
Linewidth broadening

Coupling with "bath": molecule is embedded into protein which has its own vibrational modes, acoustic modes are low-frequency almost continuous in spectrum



Mike Reppert: . *Phys. Chem. B* 2020, 124, 45, 10024–10033

Vibrations lead to ***homogeneous*** broadening, i.e. they show even when there is only one light absorbing molecule excited.

https://pubs.acs.org/doi/abs/10.1021/acs.jpcb.0c05789

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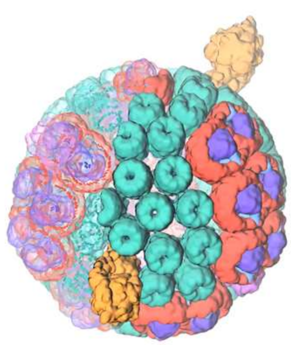
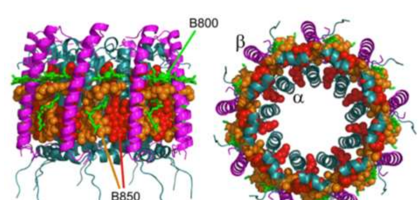
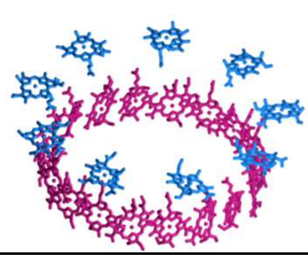
Linewidth broadening

Inhomogeneous broadening: a molecule is embedded into *different (*protein) environment that affects its electronic orbitals
Example: Bacterial Light Harvesting pigment-protein complex LH 2:





http://photobiology.info/Jones.html



Rba. sphaeroides

Melih Sener et al. https://elifesciences.org/articles/09541, https://doi.org/10.7554/eLife.09541.004

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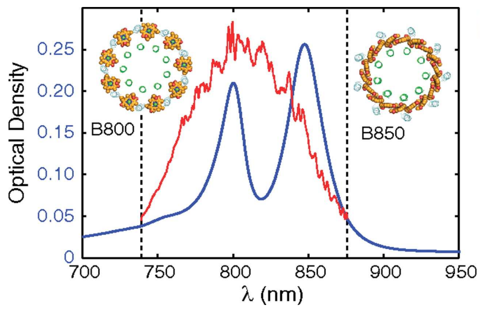
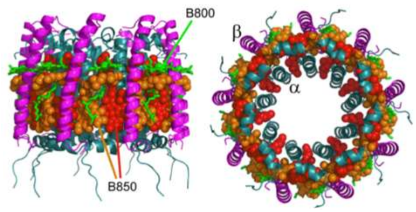
Linewidth broadening

Inhomogeneous broadening: a molecule is embedded into a *different (*protein) environment that affects its electronic orbitals
<u>Example</u>: Bacterial Light Harvesting pigment-protein complex LH 2:

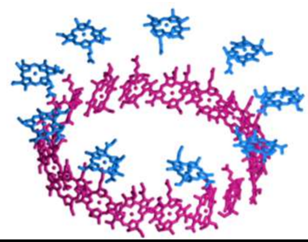
Absorption spectrum:
two bands, one from B800 BChl *a* ring, one from B850 ring – different environments.



Harel & Engel, PNAS 2012: https://www.pnas.org/content/109/3/706



http://photobiology.info/Jones.html



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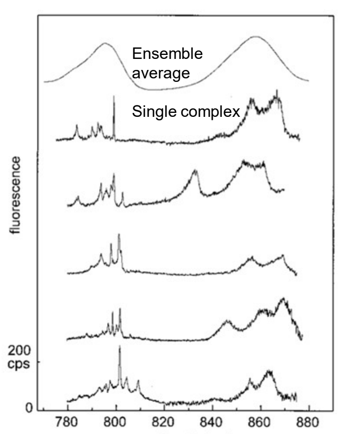
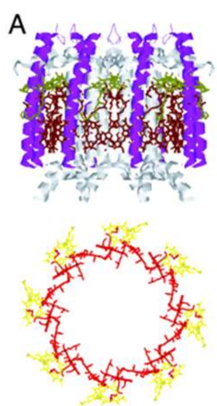
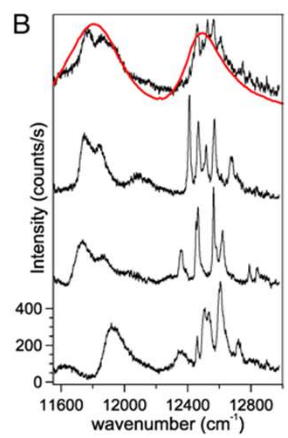
Linewidth broadening

Inhomogeneous broadening:
<u>Example</u>: Bacterial Light Harvesting pigment-protein complex LH II at T~1K.
Single-complex spectra differ from complex to complex, symmetry is lost



Oijen et al.1999 (Science)





Hoffman 2003, PNAS
https://www.pnas.org/content/100/26/15534/tab-figures-data

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Vibrational relaxation

The transition from ground state to excited state changes molecular wavefunction

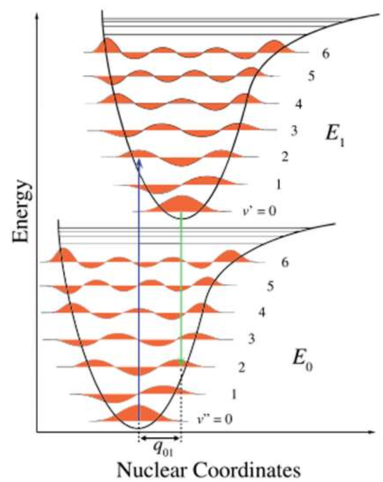
The environment (protein) structure will adapt: *vibrational relaxation*

Electronic wavefunction in an excited state must adapt to new nuclear configuration

Franck-Condon principle (approximation): Electronic transition is fast and nuclear coordinate does not change during transition.
→ Vertical transitions

Consequence: the energy of emitted photon (excited→ground) is smaller than the energy of the absorbed photon (ground→excited)

Nuclear vibrational states



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Vibrational relaxation: Stokes shift

Consequences:

- Equilibrated electronic states in upward and downward transitions are *not* the same!
- Transition energies up and down are not the same
- Transition dipole moments up and down are not the same (though usually not much different)

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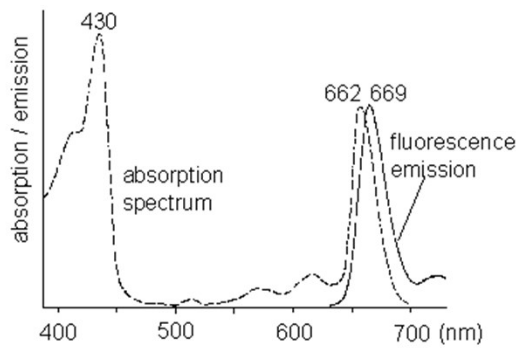
Vibrational relaxation: Stokes shift

Chlorophyll *a* absorption and emission spectra



The difference between maxima of fluorescence and emission bands is called *Stokes shift*

https://www.photosynthesis.ch/fluorescence.htm

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Photosynthesis

Important for photosynthetic efficiency:

- Homogeneous and inhomogeneous broadening:
  → more photons are absorbed from white light
  → protein can tune the energy landscape for directed (downhill) energy transfer

- Lifetime:
  → A longer lifetime allows more time for electronic energy utilization

- Transition dipole moment (TDM):
  → larger molecules can have larger TDM, absorb light more efficiently
  Chl molecule can absorb 10 photons per second under full sunlight

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